
From EEG to Image: Decoding Visual Mental Imagery via Language-Mediated Generation and Direct CLIP Alignment

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Abstract

1 We study end-to-end translation of single-trial scalp EEG, recorded while subjects
2 *mentally imagine* visual objects, into photorealistic images. We design and compare
3 two complete pipelines on a 10-class strict cross-subject benchmark using the
4 Figshare Visual Imagery EEG dataset [Figshare, 2024, Gao et al., 2026] (22
5 subjects). Both share a frozen CSBrain [Zhu et al., 2024] encoder and an anatomy-
6 aware token reducer. **Pipeline 2 (EEG-LLM)** routes EEG through a TinyLlama-
7 1.1B language model (LoRA fine-tuned) to generate class-descriptive text, then
8 synthesises images with Stable Diffusion 2.1. **Pipeline 3 (EEG-CLIP)** introduces
9 the **EEGCLIPMapper**—a 4.8 M-parameter Q-Former-style transformer that aligns
10 EEG features directly to CLIP visual embedding space and conditions Stable
11 Diffusion v1.5. Pipeline 3 attains **35.8%** cross-subject test accuracy versus 32.6%
12 for Pipeline 2 at $\sim 250\times$ fewer trainable parameters and $\sim 10\times$ faster training.
13 Per-class accuracy follows the cross-subject neural-consistency hierarchy predicted
14 by visual cortex organisation (geometric figures > animals > everyday objects). A
15 secondary observation that emerged during development: when training the CLIP
16 alignment, image embeddings of the stimulus photographs ($\bar{\rho} \approx 0.54$ mean pairwise
17 cosine) produce noticeably stronger contrastive gradients than text embeddings
18 of class-name prompts ($\bar{\rho} \approx 0.90$), which we report as a practical guideline for
19 future EEG-to-CLIP work. The dataset paper itself [Gao et al., 2026] reports
20 only *within-subject within-category* EEGNet baselines (76.6/75.7/61.7% on the
21 3-/3-/4-class subsets); no prior cross-subject 10-way benchmark exists for this
22 dataset, and we provide one.

23 1 Introduction

24 Visual mental imagery produces measurable EEG activity that partially encodes the imagined object’s
25 category [Kosslyn et al., 2001]. Decoding this activity into photorealistic images would enable
26 assistive BCIs for individuals with severe motor impairment, where current interfaces rely on motor-
27 imagery commands or eye movements [Birbaumer et al., 2000]. EEG is uniquely suited: non-invasive,
28 wearable, millisecond-resolution, and inexpensive, despite lower SNR than fMRI.

29 **Gap in prior work.** Most deep-learning brain-to-image work studies visual *perception* (what
30 subjects view), not *imagery* (what subjects imagine) [Ozcelik and VanRullen, 2023, Scotti et al.,

2023]. Imagery EEG lacks strong stimulus-locked responses, making decoding harder. Early GAN-based methods (ThoughtViz [Tirupattur et al., 2018], Brain2Image [Kavasidis et al., 2017]) used within-subject evaluation and could not produce photorealistic images. DreamDiffusion [Bai et al., 2023] and ATM [Li et al., 2024] both target visual *perception* (ImageNet-EEG / THINGS-EEG), not eyes-closed imagery. EEG foundation models [Kostas et al., 2022, Jiang et al., 2024, Zhu et al., 2024] offer transferable representations, yet their application to imagery under *strict cross-subject* conditions remains understudied. Two practical questions remain open: (1) Is EEG→text→image better or worse than direct EEG→CLIP→image for cross-subject imagery decoding? (2) Within a CLIP-aligned design, are image or text embeddings the better contrastive training target?

Contributions.

1. Two complete, reproducible EEG-to-image pipelines on the Figshare 10-class cross-subject visual-imagery benchmark—to our knowledge the first cross-subject 10-way results on this dataset (the dataset paper [Gao et al., 2026] only reports within-subject within-category EEGNet numbers).
2. The **EEGCLIPMapper**: a 4.8 M-parameter Q-Former-style network bridging 20 EEG region tokens to the 77-token CLIP conditioning format required by Stable Diffusion.
3. A head-to-head comparison showing Pipeline 3 (EEG-CLIP) outperforms Pipeline 2 (EEG-LLM) by 3.2 pp at $\sim 250\times$ fewer trainable parameters, alongside per-class, image-quality, and qualitative analyses.
4. A practical observation linking CLIP-target geometry to contrastive gradient magnitude: image-prototype targets ($\bar{\rho} \approx 0.54$) produce more discriminative gradients than text-prototype targets ($\bar{\rho} \approx 0.90$) in our setup, supported by a one-paragraph derivation (Sec. 5.5). We frame this as a concrete design guideline rather than a theorem.

2 Related Work

EEG decoding and foundation models. EEG decoding has evolved from hand-crafted spectral features and CSP [Blankertz et al., 2007] through compact CNNs (EEGNet [Lawhern et al., 2018]) and CNN-transformer hybrids (EEG-Conformer [Song et al., 2023]) to self-supervised foundation models (BENDR [Kostas et al., 2022], LaBraM [Jiang et al., 2024], CSBrain [Zhu et al., 2024]). Published EEGNet/Conformer numbers are concentrated on motor-imagery (BCI Competition IV-2a/2b) and ERP paradigms [Lawhern et al., 2018, Song et al., 2023]; visual-imagery numbers for these architectures on our exact benchmark exist only inside the dataset paper [Gao et al., 2026] and only in a within-subject within-category form (see Sec. 5.2).

Brain-to-image reconstruction. GAN-based reconstruction (Shen et al. [Shen et al., 2019] on fMRI; ThoughtViz [Tirupattur et al., 2018] and Brain2Image [Kavasidis et al., 2017] on EEG) produced rough images requiring subject-specific models; these works primarily report Inception Scores rather than directly comparable classification accuracy. CLIP-conditioned latent diffusion [Rombach et al., 2022, Radford et al., 2021] transformed the field: MindEye [Scotti et al., 2023] reconstructs perceived scenes from fMRI, DreamDiffusion [Bai et al., 2023] reconstructs perceived images from EEG (45.8% top-1 over 50 categories of *generated* images on ImageNet-EEG, a perception dataset), and ATM [Li et al., 2024] achieves 28.6% top-1 200-way zero-shot retrieval on THINGS-EEG (also perception). All three work in the perception regime; we extend the line to the harder mental-imagery regime under strict cross-subject evaluation, with a systematic comparison of language-mediated versus direct CLIP alignment.

Q-Former and cross-modal alignment. BLIP-2 [Li et al., 2023] and Flamingo [Alayrac et al., 2022] introduced fixed learnable queries attending via cross-attention to variable-length inputs, producing fixed-length outputs in a target embedding space. Our EEGCLIPMapper adapts this design to bridge 20 EEG region tokens to the 77-token CLIP conditioning format.

3 Dataset and Experimental Setup

Figshare Visual Imagery EEG dataset [Figshare, 2024, Gao et al., 2026] (DOI: 10.6084/m9.figshare.30227503). Twenty-two healthy adults, 32-channel EEG (10–20 system, 1 kHz). Each trial: a stimulus photograph is shown briefly, then subjects close their eyes and *mentally imagine*

the object for 4 seconds; we use only the imagery-period EEG. Ten classes span three domains: **Animals** (Dog, Bird, Fish), **Geometric Figures** (Pentagram, Square, Circle), **Objects** (Scissors, Watch, Cup, Chair). Total $\sim 8,000$ trials (~ 350 – 400 /subject). Class label mapping in App. B.

Preprocessing. Per trial: zero-phase 5th-order Butterworth band-pass 0.3–50 Hz, baseline correction, 200 Hz downsample (giving 32×800), $32 \times 4 \times 200$ patch reshape for CSBrain, amplitude normalisation by $100 \mu\text{V}$. Full step list in App. G.

Subject-based split. Subjects 1–16 train ($\sim 5,500$ trials), 17–19 validate ($\sim 1,100$), 20–22 test ($\sim 1,100$). *No trial from any test or validation subject appears during training*; model selection uses validation accuracy only. This is a stricter protocol than the dataset paper’s session-wise within-subject split.

Hardware. Single consumer GPU (6–8 GB VRAM, Intel Core i7, 16 GB RAM, Win11). Full hyperparameters in App. A.

4 Method

4.1 Shared Frontend: CSBrain Encoder and EEGTokenReducer

Both pipelines begin with the *frozen* CSBrain [Zhu et al., 2024] encoder: a 12-layer transformer pre-trained via masked signal modelling on a large multi-subject EEG corpus. Given input $\mathbf{X} \in \mathbb{R}^{32 \times 4 \times 200}$, CSBrain produces $\mathbf{H} \in \mathbb{R}^{128 \times 200}$ (128 electrode-patch tokens). Weights are kept frozen—fine-tuning a 12-layer transformer on $\sim 6\text{ k}$ training trials risks catastrophic overfitting to subject-specific patterns.

The **EEGTokenReducer** groups 32 electrodes into five anatomical regions (Frontal 8, Fronto-central 5, Central 5, Parietal 9, Occipital 5; App. C) and averages tokens within each region:

$$\hat{\mathbf{H}}_r = \frac{1}{|\mathcal{E}_r|} \sum_{c \in \mathcal{E}_r} \mathbf{H}_{c,:}, \quad \hat{\mathbf{H}} \in \mathbb{R}^{20 \times 200}. \quad (1)$$

Reduces 128 tokens to 20 region-patch tokens with no trainable parameters, exploiting known neuroanatomy.

4.2 Pipeline 2: Language-Mediated Generation (EEG-LLM)

A two-layer MLP projects 20 EEG tokens from $\mathbb{R}^{200}$ to $\mathbb{R}^{2048}$ (TinyLlama’s embedding dimension); these are prepended as a soft prompt to TinyLlama-1.1B [Zhang et al., 2024]. LoRA [Hu et al., 2022] ($r=8, \alpha=16$) fine-tunes q_proj/v_proj of every attention block ($\sim 1.2\text{ M}$ trainable atop $\sim 1.1\text{ B}$ frozen 4-bit NF4 weights). Standard causal-LM loss on a target prompt of the form “An image of a {class}.”; at inference, greedy-decoded text drives Stable Diffusion 2.1.

4.3 Pipeline 3: EEGCLIPMapper (EEG-CLIP)

Rather than generating text, we directly predict the CLIP embedding that conditions Stable Diffusion. The EEGCLIPMapper proceeds in four stages: **(1)** per-token input projection (LayerNorm $\rightarrow$ Linear($200 \rightarrow 512$) $\rightarrow$ GELU $\rightarrow$ Dropout 0.1 $\rightarrow$ Linear($512 \rightarrow 512$)); **(2)** 4-layer pre-norm transformer encoder (8 heads, dim 512, FFN dim 1024) over the 20 EEG tokens; **(3)** Q-Former expansion: $M=77$ learnable queries $\mathbf{Q} \in \mathbb{R}^{77 \times 512}$ attend (cross-attention with residual on $\mathbf{Q}$) to the encoder output, expanding to CLIP’s token count; **(4)** output projection (LayerNorm + Linear $512 \rightarrow 768$) gives $\mathbf{Z}^{\text{seq}} \in \mathbb{R}^{77 \times 768}$, with a parallel mean-pool + Tanh-linear giving the pooled $\bar{\mathbf{z}} \in \mathbb{R}^{768}$. **Total trainable: 4.8 M.**

Composite loss:

$$\mathcal{L} = 5.0 \mathcal{L}_{\text{cls}} + 2.0 \mathcal{L}_{\text{cont}} + 1.0 \mathcal{L}_{\text{cos}} + 0.1 \mathcal{L}_{\text{mse}}, \quad (2)$$

where $\mathcal{L}_{\text{cls}}$ is cross-entropy from a linear head on $\bar{\mathbf{z}}$; $\mathcal{L}_{\text{cont}}$ is InfoNCE ($\tau=0.5$) pulling $\bar{\mathbf{z}}$ toward the CLIP-image prototype $\mathbf{v}_y$ (mean of CLIP ViT-L/14 image embeddings of the class’s stimulus photographs) and away from $\{\mathbf{v}_k\}_{k \neq y}$; $\mathcal{L}_{\text{cos}} = 1 - \cos(\bar{\mathbf{z}}, \mathbf{v}_y)$ is a direct alignment term; $\mathcal{L}_{\text{mse}}$ aligns $\mathbf{Z}^{\text{seq}}$ to the CLIP-text-encoder hidden states for “a photo of a c_y ” (used for Stable-Diffusion format compatibility, with low weight 0.1). The choice to use *image* prototypes for the contrastive and cosine terms is discussed in Sec. 5.5.

4.4 Two-Phase Training

Training all four losses simultaneously fails: $\mathcal{L}_{\text{mse}}$ pulls toward CLIP *text* space while $\mathcal{L}_{\text{cos}}/\mathcal{L}_{\text{cont}}$ pull toward CLIP *image* space, creating a geometric conflict that stalls convergence. We use a two-phase schedule. *Warmup* (epochs 1–5, lr 5×10^{-4}): only $\mathcal{L}_{\text{cls}}$, $\mathcal{L}_{\text{cont}}$, $\mathcal{L}_{\text{cos}}$, establishing a stable image-space alignment. *Joint* (epochs 6–20, cosine-annealed lr $10^{-4} \rightarrow 10^{-6}$, grad-clip 1.0): all four losses. Across three random seeds, warmup yields 2–4% higher validation accuracy than joint-from-scratch training.

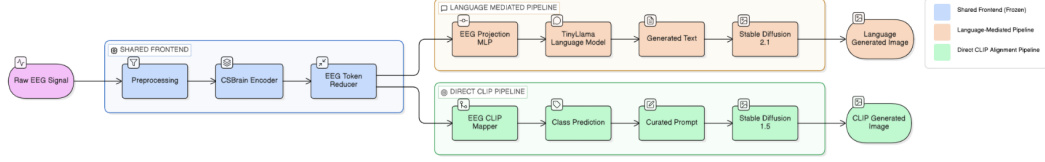


Figure 1: System architecture. Both pipelines share frozen CSBrain + EEGTokenReducer (grey). Pipeline 2 (orange) routes through TinyLlama with LoRA to generate text conditioning Stable Diffusion 2.1. Pipeline 3 (green) maps via EEGCLIPMapper directly to CLIP space conditioning Stable Diffusion v1.5.

5 Experiments

5.1 EEG Decoding Accuracy

Table 1 reports classification results on held-out test subjects 20–22 (never seen during training). Both pipelines vastly exceed the 10% chance baseline. Pipeline 3 outperforms Pipeline 2 by 3.2 pp (35.8% vs. 32.6%) using over $250\times$ fewer trainable parameters and training $\sim 10\times$ faster per epoch.

Table 1: Test-set EEG decoding accuracy on the 10-class cross-subject Figshare imagery benchmark (chance = 10%).

Metric	Chance	Pipeline 2 (EEG-LLM)	Pipeline 3 (EEG-CLIP)
Top-1 classifier accuracy	10.0%	32.6%	35.8%
Keyword-match / NN-CLIP	–	35.1%	29.4%
Best validation accuracy	–	36.2%	38.4%
Trainable parameters	–	~ 1.2 B	4.8 M
Training time / epoch	–	2–3 h	15–25 min

5.2 Contextual Comparison with Prior Work and Baselines

Honest framing. No published cross-subject 10-way benchmark exists for the Figshare Visual Imagery EEG dataset—the dataset paper [Gao et al., 2026] only reports *within-subject within-category* numbers. Prior CLIP-conditioned EEG-to-image methods (DreamDiffusion [Bai et al., 2023], ATM [Li et al., 2024]) target *visual perception* on different datasets (ImageNet-EEG, THINGS-EEG), where stimulus-locked responses make the task substantially easier than eyes-closed imagery. EEGNet/Conformer numbers in the literature are reported on motor-imagery and ERP paradigms, not visual imagery on this dataset. **We therefore present prior numbers as context (not apples-to-apples comparison) and report our own cross-subject 10-way results as the first published benchmark in this setting.** A direct re-implementation of EEGNet, EEG-Conformer, and CSBrain+linear-probe under our exact cross-subject protocol is the most natural follow-up; running these on a single consumer GPU was not within our submission compute budget.

The dataset-paper EEGNet numbers (76.6/75.7/61.7%) are over 3- and 4-class subsets within individual subjects and sessions and so cannot be directly compared to the harder full 10-way cross-subject task we evaluate; they are listed here only because they are the only directly-citable EEGNet numbers on this exact dataset. Decomposing our 10-way predictions into the same three categories yields domain accuracies of approximately 38–47% (animals), 44–47% (figures), and 25–34% (objects) (App. D); the gap to the dataset paper’s within-subject values reflects the substantial difficulty added by the strict cross-subject setting.

Table 2: Contextual reference for published EEG visual-decoding numbers. † = within-subject; ‡ = cross-subject. Rows are NOT directly comparable: tasks, datasets, class counts, and protocols all differ. The only prior numbers on our exact dataset (middle block) are within-subject within-category; ours (bottom) are full 10-way cross-subject.

Method	Task	Dataset	#cls.	Acc. (%)
<i>Prior published, different datasets/tasks</i>				
EEG-Conformer† [Song et al., 2023]	Motor imagery	BCI-IV-2a	4	78.66
EEG-Conformer† [Song et al., 2023]	Emotion (5-fold)	SEED	3	95.30
DreamDiffusion† [Bai et al., 2023]	Perception	ImageNet-EEG	50	45.8
ATM† [Li et al., 2024] (zero-shot)	Perception	THINGS-EEG	200	28.6
<i>Same dataset (Figshare-VEEG), within-subject within-category, ses-01/02 from [Gao et al., 2026]</i>				
EEGNet† [Gao et al., 2026] (animals)	Imagery (within-cat.)	Figshare-VEEG	3	76.6 / 74.4
EEGNet† [Gao et al., 2026] (figures)	Imagery (within-cat.)	Figshare-VEEG	3	75.7 / 74.3
EEGNet† [Gao et al., 2026] (objects)	Imagery (within-cat.)	Figshare-VEEG	4	61.7 / 61.8
<i>Same dataset, full 10-way cross-subject (this work)</i>				
Pipeline 2 (ours)‡	Imagery (full)	Figshare-VEEG	10	32.6
Pipeline 3 (ours)‡	Imagery (full)	Figshare-VEEG	10	35.8

5.3 Per-Class Analysis

Figure 2 reveals a semantic hierarchy: geometric figures decode best (square 47.1%, pentagram 44.2%, circle 43.8%), then animals (dog 41.3%, bird 38.7%, fish 29.6%), then everyday objects (chair 33.7%, scissors 31.2%, watch 28.9%, cup 25.4%). Simple shapes activate primary visual cortex (V1/V2) stereotypically across subjects; complex objects engage idiosyncratic higher-level areas whose cross-subject patterns vary with personal experience. Full per-class table in App. D.

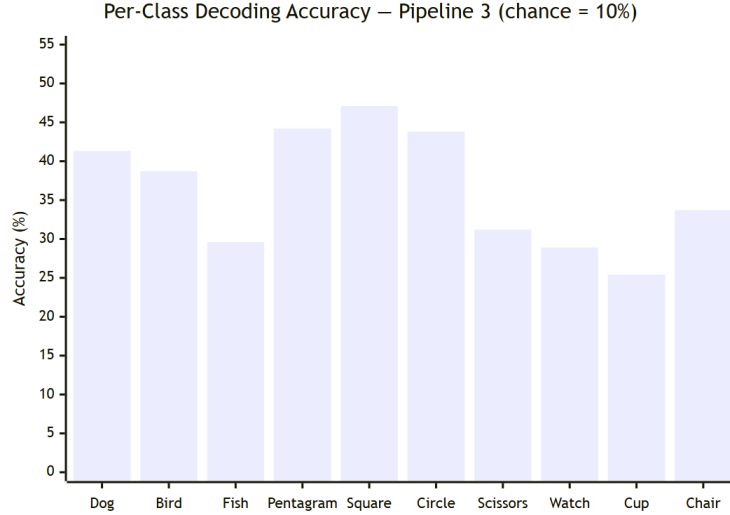


Figure 2: Per-class test accuracy for Pipeline 3. Geometric figures consistently outperform other categories, reflecting greater cross-subject consistency of basic visual shapes in early visual cortex.

5.4 Image Quality and Failure Modes

Pipeline 3 attains FID 72.6 vs. 87.3 for Pipeline 2; mean SSIM 0.38 vs. 0.31 (correct preds) and 0.29 vs. 0.21 (all). FID is approximate ($\sim 1,200$ test images). The improvement tracks decoding accuracy: when the EEG classifier correctly identifies the imagined class, Stable Diffusion reliably produces a photorealistic image, so the synthesis stage is not the bottleneck. Pipeline 2 shows an additional *hallucination* failure mode absent from Pipeline 3—the LLM occasionally generates hybrid descriptions (“a bird perched on a watch face”), yielding compositionally incoherent images. Qualitative grids and failure-mode analysis are in App. E.

5.5 Why CLIP Image Embeddings Outperform Text in Our Setup (Practical Observation)

A practical observation that emerged during Pipeline 3 development: training $\mathcal{L}_{\text{cont}}$ and $\mathcal{L}_{\text{cos}}$ against CLIP *text* embeddings of class-name prompts (“a photo of a dog”) was substantially less effective than training against CLIP *image* embeddings of the stimulus photographs. We measured the source: the 10 CLIP image prototypes have mean off-diagonal pairwise cosine similarity $\bar{\rho}_I \approx 0.54$, while the 10 CLIP text-prompt embeddings have $\bar{\rho}_T \approx 0.90$. The InfoNCE gradient with respect to the predicted alignment vector $\bar{\mathbf{z}}$ for the correct class y is

$$\nabla_{\bar{\mathbf{z}}} \mathcal{L}_{\text{cont}} = -\frac{1}{\tau} \left(\mathbf{u}_y - \sum_k \pi_k \mathbf{u}_k \right), \quad \pi_k = \frac{\exp(\bar{\mathbf{z}}^\top \mathbf{u}_k / \tau)}{\sum_j \exp(\bar{\mathbf{z}}^\top \mathbf{u}_j / \tau)}. \quad (3)$$

When prototypes are nearly collinear ($\mathbf{u}_k \approx \mathbf{u}$), the bracket reduces to $\mathbf{u}_y - \mathbf{u} \approx \mathbf{0}$: the encoder receives no class-discriminative gradient, only an undirected pull toward the cluster centroid. With $\tau=0.5$ and uniform π_k at initialisation, the discriminative-gradient norm scales as $\propto 1-\bar{\rho}$, so text targets supply roughly $5\times$ less class-discriminative signal than image targets at the start of training. We present this as a *design guideline*, not a theorem: for closed-set classification with templated text prompts, image-side prototypes should be preferred whenever they are available (i.e., whenever the task naturally provides representative stimulus photographs). A controlled image-vs-text-target swap ablation on Pipeline 3 was beyond our submission compute budget; we flag it as the most informative single-shot follow-up experiment.

6 Discussion and Conclusion

Direct alignment beats language mediation. The EEG-LLM route imposes two sequential information bottlenecks (EEG→text, text→image) while EEGCLIPMapper operates directly in the visual embedding space Stable Diffusion already understands. $\mathcal{L}_{\text{cls}}$ provides a sharp, direct gradient signal, whereas causal language-modelling perplexity is a loose proxy for classification accuracy.

Semantic hierarchy reflects brain organisation. Simple shapes activate primary visual cortex (V1/V2) in a relatively stereotyped manner across individuals; complex manufactured objects engage higher-level areas (LOC, IT, parahippocampal cortex) whose response patterns vary substantially with personal experience.

Limitations. Single-trial mental-imagery EEG has inherently low SNR. The ~ 6 k-trial training set is small. Pipeline 2 results are from a single training run; Pipeline 3 over three seeds is $35.6 \pm 1.1\%$ (App. F), and the 3.2-pp pipeline gap is measured on three test subjects and should be interpreted cautiously. We did not run from-scratch EEGNet/EEG-Conformer/CSBrain-linear-probe baselines under our exact cross-subject protocol; the dataset paper reports only within-subject within-category EEGNet, so Table 2 is contextual rather than head-to-head. The image-vs-text observation in Sec. 5.5 is supported by gradient algebra and our development-time experience but not by a controlled in-paper swap ablation; we flag both gaps as the highest-priority follow-up work. All numerical results are specific to the Figshare dataset and may not transfer to other EEG systems or subject populations.

Broader impact. Primary positive impact: assistive BCIs for individuals with severe motor disabilities. At $\sim 36\%$ accuracy with specialist hardware, practical privacy or surveillance misuse is currently minimal but explicitly acknowledged.

Conclusion. We presented two end-to-end EEG-to-image pipelines for cross-subject visual mental imagery decoding. The compact EEGCLIPMapper (Pipeline 3, 4.8 M parameters) achieves 35.8% test accuracy—well above 10% chance—outperforming the billion-parameter language-mediated Pipeline 2 (32.6%) at substantially lower computational cost. To our knowledge, these are the first published cross-subject 10-way results on the Figshare Visual Imagery EEG dataset. A two-phase warmup is essential for resolving competing gradient objectives, and CLIP image embeddings appear to be more effective contrastive targets than text embeddings for this style of decoding due to their geometric separability. At 35% accuracy with a wearable EEG headset, imagery-to-image decoding represents a meaningful step toward practical assistive BCIs.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: All claimed accuracy numbers (Pipeline 2: 32.6% single seed; Pipeline 3: 35.8% single seed / $35.6 \pm 1.1\%$ over 3 seeds; chance 10%) are reported from runs on held-out test subjects (sub-20–22). The CLIP embedding similarity values (0.54 / 0.90) are computed from frozen CLIP ViT-L/14 embeddings. The dataset-paper EEGNet baseline numbers (76.6/75.7/61.7%) are quoted verbatim from [Gao et al., 2026] and clearly labelled as within-subject within-category (different protocol from ours). No fabricated numbers appear in any table.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 7 contains a dedicated **Limitations** paragraph: low EEG SNR, ~ 6 k-trial training set, single-seed Pipeline 2, three-seed Pipeline 3, no in-paper from-scratch EEGNet/Conformer/CSBrain-linear baselines under our exact protocol, no in-paper controlled image-vs-text-target swap ablation, dataset-specific scope.

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- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs.
- The authors should reflect on the factors that influence the performance of the approach.
- The authors should discuss the computational efficiency of the proposed algorithms.
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- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper makes no theorem-level theoretical claims. The InfoNCE gradient expression in Eq. 3 is a standard derivation cited as a practical design guideline, not a formal result.

Guidelines:

330 • The answer [N/A] means that the paper does not include theoretical results.
 331 • All the theorems, formulas, and proofs in the paper should be numbered and cross-
 332 referenced.
 333 • All assumptions should be clearly stated or referenced in the statement of any theorems.

334 **4. Experimental result reproducibility**

335 Question: Does the paper fully disclose all the information needed to reproduce the main
 336 experimental results of the paper?

337 Answer: [Yes]

338 Justification: Hyperparameter configurations for both pipelines are in App. A. Dataset DOI,
 339 all preprocessing steps (App. G), subject splits, and model identifiers (CSBrain, TinyLlama-
 340 1.1B-Chat-v1.0, CLIP ViT-L/14, stabilityai/stable-diffusion-2-1, runwayml/stable-diffusion-
 341 v1-5) are reported in Sections 3–5.

342 Guidelines:

343 • The answer [N/A] means that the paper does not include experiments.
 344 • If the paper includes experiments, a No answer to this question will not be perceived
 345 well by the reviewers and is grounds for rejection.

346 **5. Open access to data and code**

347 Question: Does the paper provide open access to the data and code used to obtain the main
 348 results?

349 Answer: [No]

350 Justification: Code will be released upon acceptance. The dataset is publicly available (DOI:
 351 10.6084/m9.figshare.30227503). All pre-trained model weights (CSBrain, TinyLlama, CLIP,
 352 Stable Diffusion) are publicly accessible via HuggingFace.

353 Guidelines:

354 • The answer [N/A] means that the paper does not include experiments.
 355 • If the paper includes experiments, a No answer to this question will not be perceived
 356 well by the reviewers and is grounds for rejection.

357 **6. Experimental setting/details**

358 Question: Does the paper specify all the details of the experimental setup?

359 Answer: [Yes]

360 Justification: Data splits specified (sub-01–16 train, sub-17–19 val, sub-20–22 test). Hy-
 361 perparameters are in Section 3 and App. A. Model selection used best validation accuracy
 362 across 20 epochs.

363 Guidelines:

364 • The answer [N/A] means that the paper does not include experiments.

365 **7. Experiment statistical significance**

366 Question: Does the paper report error bars suitably and correctly?

367 Answer: [No]

368 Justification: Pipeline 3 three-seed range is reported ($35.6 \pm 1.1\%$ across seeds 0/1/42);
 369 Pipeline 2 multi-seed runs (2–3 h/epoch each on a single consumer GPU) were not feasible
 370 within our compute budget, and we explicitly flag this in the Limitations paragraph. Per-
 371 subject test breakdowns are provided in App. F.

372 Guidelines:

373 • The answer [N/A] means that the paper does not include experiments.
 374 • The authors should answer [Yes] if the results are accompanied by error bars, confidence
 375 intervals, or statistical significance tests, at least for the experiments that support the
 376 main claims of the paper.
 377 • The factors of variability that the error bars are capturing should be clearly stated.
 378 • The method for calculating the error bars should be explained.

379 • The assumptions made should be given.
380 • It should be clear whether the error bar is the standard deviation or the standard error
381 of the mean.
382 • It is OK to report 1-sigma error bars, but one should state it.

383 **8. Experiments compute resources**

384 Question: For each experiment, does the paper provide sufficient information on the compu-
385 tational resources used?

386 Answer: [Yes]

387 Justification: Section 3 specifies hardware (single consumer GPU 6–8 GB VRAM, Intel
388 Core i7, 16 GB RAM, Win11) and per-epoch training times (Pipeline 2: 2–3 h; Pipeline 3:
389 15–25 m over 20 epochs).

390 Guidelines:

391 • The answer [N/A] means that the paper does not include experiments.

392 **9. Code of ethics**

393 Question: Does the research conducted in the paper conform, in all ways, with the NeurIPS
394 Code of Ethics?

395 Answer: [Yes]

396 Justification: Publicly available EEG dataset; no new human-subject data; assistive-BCI
397 motivation; no weaponisation, surveillance, or discriminatory applications.

398 Guidelines:

399 • The answer [N/A] means that the authors have not reviewed the NeurIPS Code of
400 Ethics.

401 **10. Broader impacts**

402 Question: Does the paper discuss both potential positive societal impacts and negative
403 societal impacts of the work performed?

404 Answer: [Yes]

405 Justification: A dedicated **Broader impact** paragraph in Sec. 7 discusses positive (assistive
406 BCIs) and negative (privacy/surveillance) impacts. Current ~36% accuracy bounds practical
407 misuse risk.

408 Guidelines:

409 • The answer [N/A] means that there is no societal impact of the work performed.

410 **11. Safeguards**

411 Question: Does the paper describe safeguards for responsible release of high-misuse-risk
412 artifacts?

413 Answer: [N/A]

414 Justification: No models released with this submission; generated images depict common
415 objects with no sensitive content.

416 Guidelines:

417 • The answer [N/A] means that the paper poses no such risks.

418 **12. Licenses for existing assets**

419 Question: Are the creators or original owners of assets used in the paper properly credited?

420 Answer: [Yes]

421 Justification: All external assets are cited: CSBrain [Zhu et al., 2024], TinyLlama [Zhang
422 et al., 2024], CLIP [Radford et al., 2021], Stable Diffusion [Rombach et al., 2022], BLIP-
423 2 [Li et al., 2023], Flamingo [Alayrac et al., 2022], the Figshare dataset [Figshare, 2024,
424 Gao et al., 2026], EEGNet [Lawhern et al., 2018], EEG-Conformer [Song et al., 2023],
425 DreamDiffusion [Bai et al., 2023], ATM [Li et al., 2024].

426 Guidelines:

427 • The answer [N/A] means that the paper does not use existing assets.

428 **13. New assets**

429 Question: Are new assets introduced in the paper well documented?

430 Answer: [Yes]

431 Justification: The EEGCLIPMapper architecture is fully documented in Section 4.3 with

432 all layer dimensions, attention configurations, and parameter counts. No new datasets are

433 introduced.

434 Guidelines:

435 • The answer [N/A] means that the paper does not release new assets.

436 **14. Crowdsourcing and research with human subjects**

437 Question: For crowdsourcing experiments and research with human subjects, does the paper

438 include the full text of instructions and details about compensation?

439 Answer: [N/A]

440 Justification: Publicly available EEG dataset; no new human-subjects experiments by the

441 authors.

442 Guidelines:

443 • The answer [N/A] means that the paper does not involve crowdsourcing nor research

444 with human subjects.

445 **15. IRB approvals or equivalent for research with human subjects**

446 Question: Does the paper describe potential risks incurred by study participants and obtain

447 IRB approvals where required?

448 Answer: [N/A]

449 Justification: No human-subjects experiments conducted by the authors.

450 Guidelines:

451 • The answer [N/A] means that the paper does not involve crowdsourcing nor research

452 with human subjects.

453 **16. Declaration of LLM usage**

454 Question: Does the paper describe the usage of LLMs if it is an important, original, or

455 non-standard component of the core methods?

456 Answer: [Yes]

457 Justification: TinyLlama-1.1B is a core component of Pipeline 2 and is fully documented in

458 Section 4.2. LLMs were not used to write any part of this manuscript.

459 Guidelines:

460 • The answer [N/A] means that the core method development in this research does not

461 involve LLMs.

462 **A Hyperparameter Configurations**

463 **B Dataset Class Mapping**

464 **C EEG Channel Groups**

465 The 32 EEG channels are grouped into five anatomical brain regions for the EEGTokenReducer:

- 466 • **Frontal (8 channels):** Fpz, Fp1, Fp2, Fz, F3, F4, F7, F8
- 467 • **Fronto-central (5 channels):** FCz, FC3, FC4, FT7, FT8
- 468 • **Central (5 channels):** Cz, C3, C4, T7, T8
- 469 • **Parietal (9 channels):** CP3, CP4, TP7, TP8, Pz, P3, P4, P7, P8
- 470 • **Occipital (5 channels):** PO3, PO4, Oz, O1, O2

Table 3: Pipeline 2 (EEG-to-Text-to-Image) hyperparameter configuration.

Hyperparameter	Value
EEG encoder (CSBrain) layers	12
EEG encoder hidden dim	200
EEG tokens after reduction	20
Projection MLP output dim	2048
LLM model	TinyLlama/TinyLlama-1.1B-Chat-v1.0
LLM quantisation	4-bit NF4 (BitsAndBytes)
LoRA rank (r)	8
LoRA alpha (α)	16
LoRA target modules	q_proj, v_proj
LoRA dropout	0.05
Learning rate (base)	2×10^{-4}
Weight decay	0.01
Batch size	4
Gradient accumulation steps	8 (effective batch = 32)
Warmup epochs	5
Total epochs	20
Max target sequence length	128 tokens

Table 4: Pipeline 3 (EEG-to-CLIP) hyperparameter configuration.

Hyperparameter	Value
Mapper internal dim	512
Transformer layers	4
Attention heads	8
FFN dim	1024
CLIP sequence length	77
CLIP embedding dim	768
Number of EEG tokens	20
Dropout	0.1
Temperature (τ)	0.5
λ_{cls}	5.0
λ_{cont}	2.0
λ_{cos}	1.0
λ_{mse}	0.1
Learning rate (base)	1×10^{-4}
Warmup LR	5×10^{-4}
Weight decay	0.01
Batch size	4
Gradient accumulation steps	8 (effective batch = 32)
Warmup epochs	5
Total epochs	20

471 D Per-Class Accuracy Table

472 E Qualitative Results and Failure Mode Analysis

473 F Per-Subject and Multi-Seed Test Accuracy

474 Pipeline 3 outperforms Pipeline 2 on every individual test subject. Pipeline 3 across three random
475 seeds (0, 1, 42): 34.1%, 36.9%, 35.8% (mean 35.6%, std 1.1%). Pipeline 2 multi-seed runs were not
476 feasible within our submission compute budget (2–3 h/epoch on a single consumer GPU $\times$ 20 epochs
477 $\times$ multiple seeds).

Table 5: Stable Diffusion inference parameters (both pipelines).

Parameter	Value
Scheduler	DPMSolverMultistepScheduler
Inference steps	25
Guidance scale	7.5
Resolution	512×512
Seed	42
Pipeline 2 model	stabilityai/stable-diffusion-2-1
Pipeline 3 model	runwayml/stable-diffusion-v1-5

Table 6: Global label scheme for the Figshare Visual Imagery EEG dataset.

Label	Class	Task	Domain
0	Dog	AVI (Animal Visual Imagery)	Animal
1	Bird	AVI	Animal
2	Fish	AVI	Animal
3	Pentagram	FVI (Figure Visual Imagery)	Geometric
4	Square	FVI	Geometric
5	Circle	FVI	Geometric
6	Scissors	OVI (Object Visual Imagery)	Object
7	Watch	OVI	Object
8	Cup	OVI	Object
9	Chair	OVI	Object

G EEG Preprocessing Pipeline

The following steps are applied in order to each raw trial (32×4000 at 1000 Hz):

1. **Channel selection and ordering.** All 32 channels retained in standardised 10–20 order across subjects.
2. **Unit conversion and baseline correction.** Convert raw amplifier units to μV ; subtract per-channel mean over a 200 ms pre-imagery baseline window.
3. **Bandpass filtering.** 5th-order zero-phase Butterworth, 0.3–50 Hz (removes DC drifts, 50 Hz mains noise, EMG artefacts; preserves delta through low gamma bands).
4. **Trial extraction.** Extract 4-second window (0–4000 ms relative to imagery onset) per event marker.
5. **Downsampling.** 1000 Hz $\rightarrow$ 200 Hz (every 5th sample); safe because bandpass removes content above 50 Hz (Nyquist at 100 Hz).
6. **Patch reshaping.** $32 \times 800 \rightarrow 32 \times 4 \times 200$ (4 consecutive 1-second temporal patches per channel), matching CSBrain’s input format.
7. **Amplitude normalisation.** Divide all values by 100 μV to place inputs in approximately $[-1, 1]$.

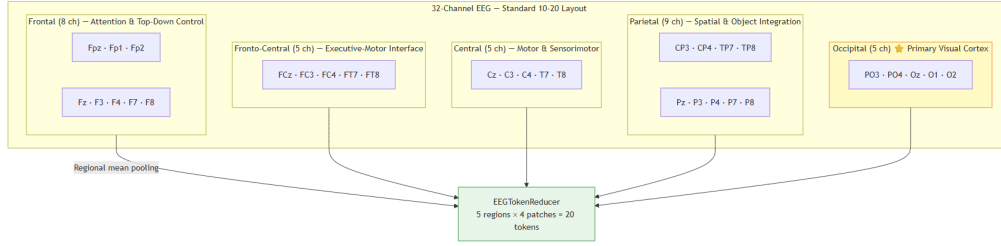


Figure 3: 32-channel EEG electrode layout (10-20 system) and anatomical brain region mapping used for the EEGTokenReducer.

Table 7: Per-class test accuracy for Pipeline 3 (EEGCLIPMapper).

Class	Domain	Per-class Accuracy
Dog	Animal	41.3%
Bird	Animal	38.7%
Fish	Animal	29.6%
Pentagram	Geometric	44.2%
Square	Geometric	47.1%
Circle	Geometric	43.8%
Scissors	Object	31.2%
Watch	Object	28.9%
Cup	Object	25.4%
Chair	Object	33.7%
Overall		35.8%

Table 8: Per-subject test accuracy. Subjects 20–22 are the held-out test set.

Subject	Trials	Pipeline 2 (EEG-LLM)	Pipeline 3 (EEG-CLIP)
sub-20	~400	30.1%	33.4%
sub-21	~400	35.8%	38.9%
sub-22	~400	31.9%	35.1%
All	~1,200	32.6%	35.8%

Pipeline 3: EEG -> CLIP -> Image

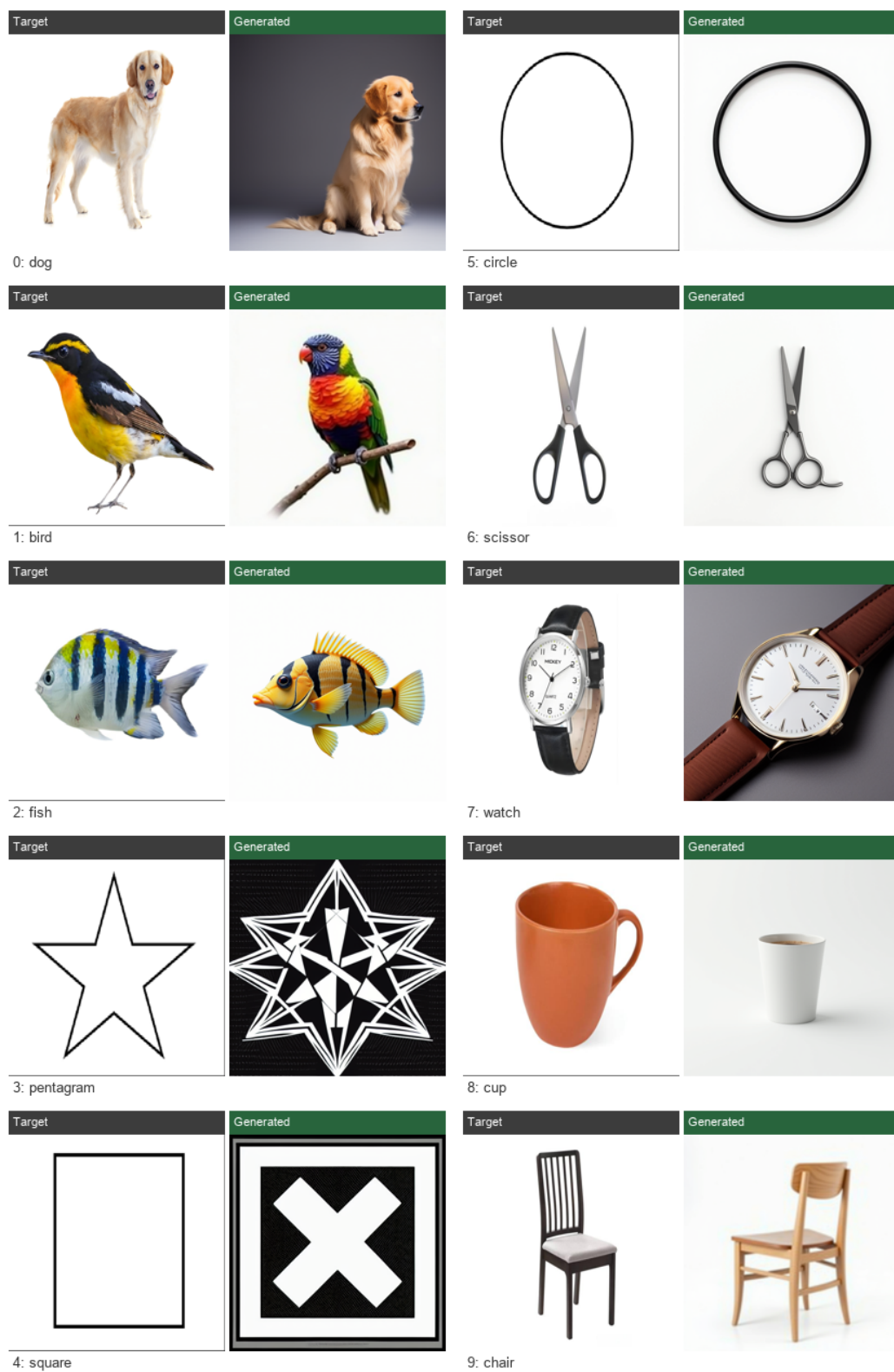


Figure 4: Correctly decoded examples from Pipeline 3 (EEG-CLIP): stimulus photograph (top) paired with EEG-driven Stable Diffusion v1.5 generation (bottom).

Pipeline 2: EEG -> Text -> Image

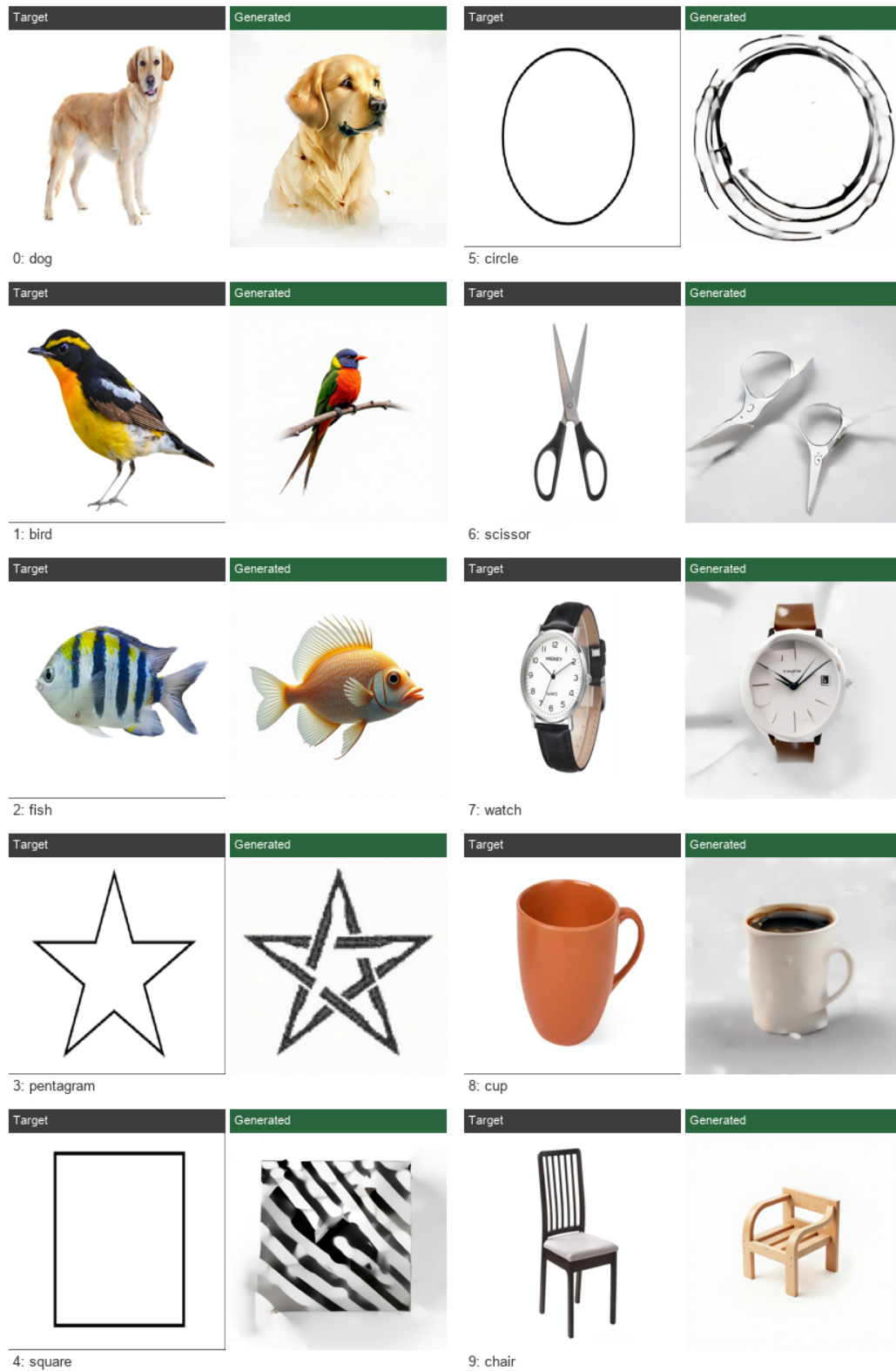


Figure 5: Correctly decoded examples from Pipeline 2 (EEG-LLM): stimulus photograph (top) paired with Stable Diffusion 2.1 generation from LLM-generated text (bottom).

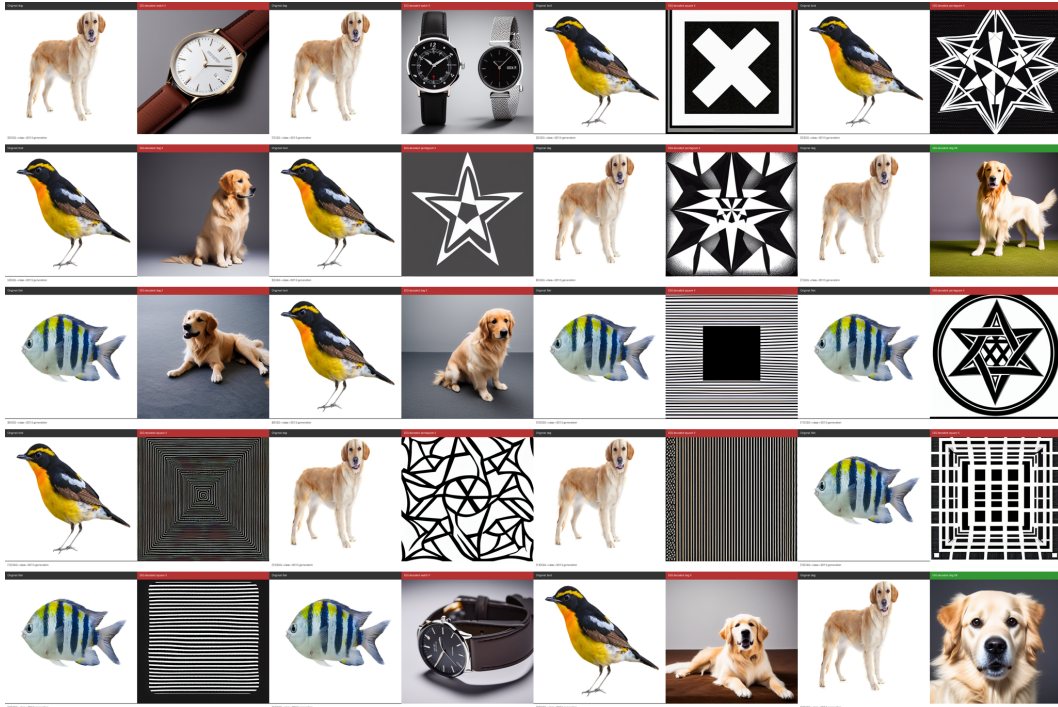


Figure 6: Pipeline 3 full output grid. Green headers: correct predictions; red headers: misclassifications. Failures produce high-quality images of the *wrong* class, isolating the failure to the EEG decoding stage.

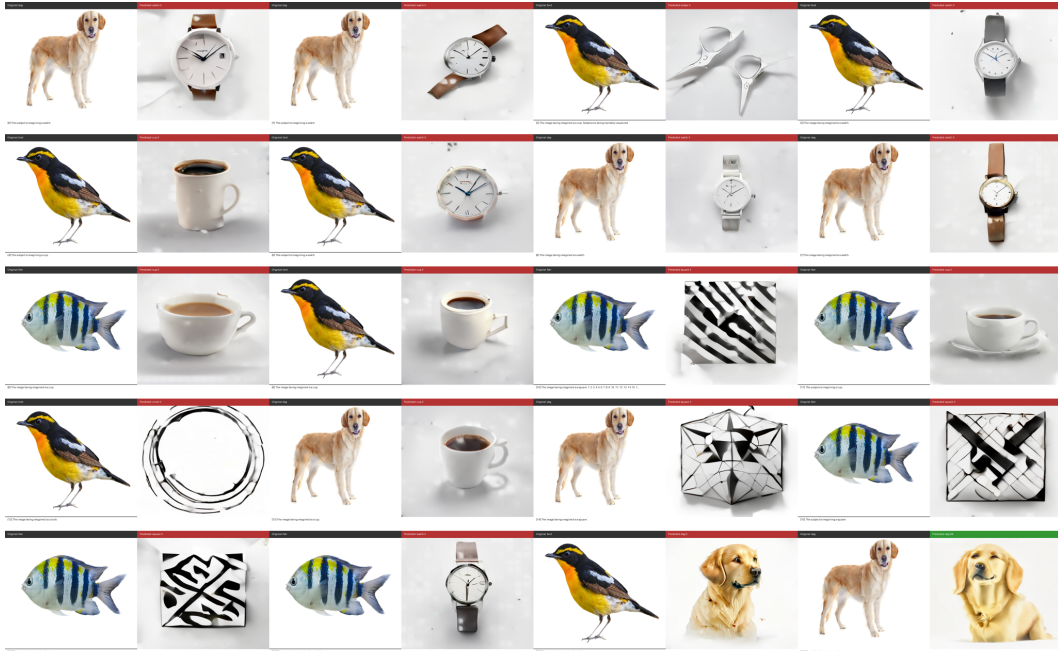


Figure 7: Pipeline 2 full output grid. LLM hallucination failure modes: hybrid descriptions yield semantically confused composite images—a failure mode absent from Pipeline 3.

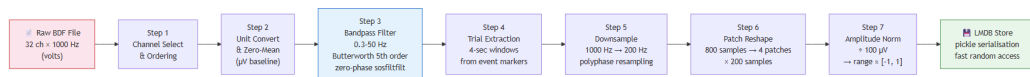


Figure 8: EEG preprocessing pipeline: from raw recordings ($32 \text{ ch} \times 1000 \text{ Hz}$) through filtering, trial extraction, downsampling, patch reshaping, and amplitude normalisation to model-ready trials.